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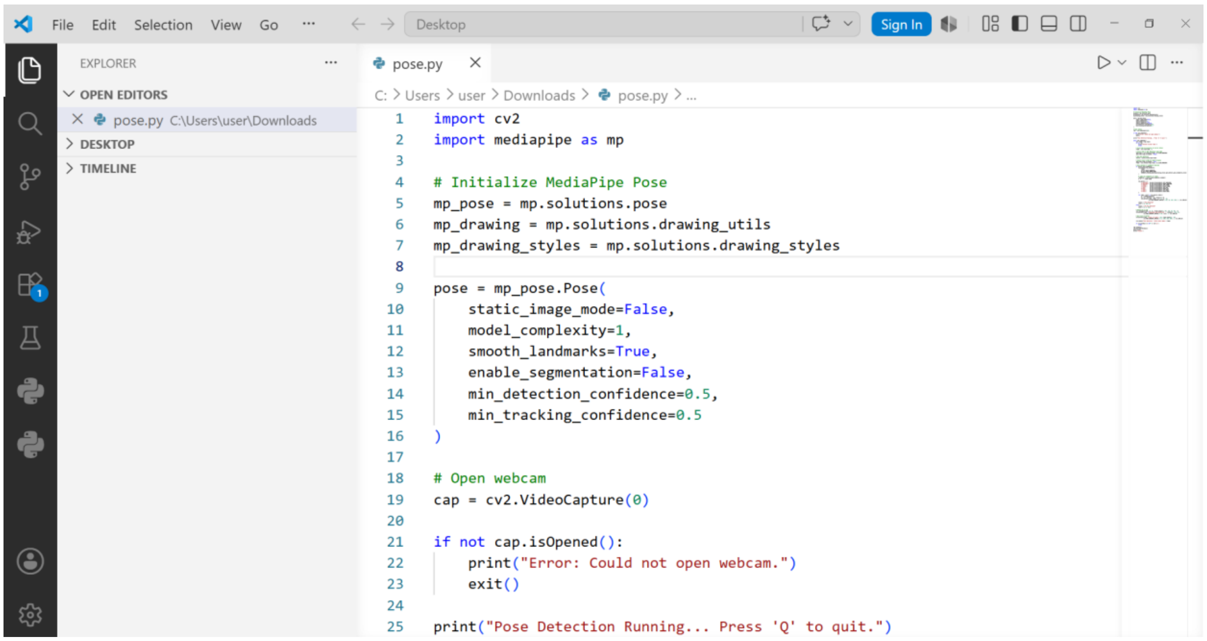
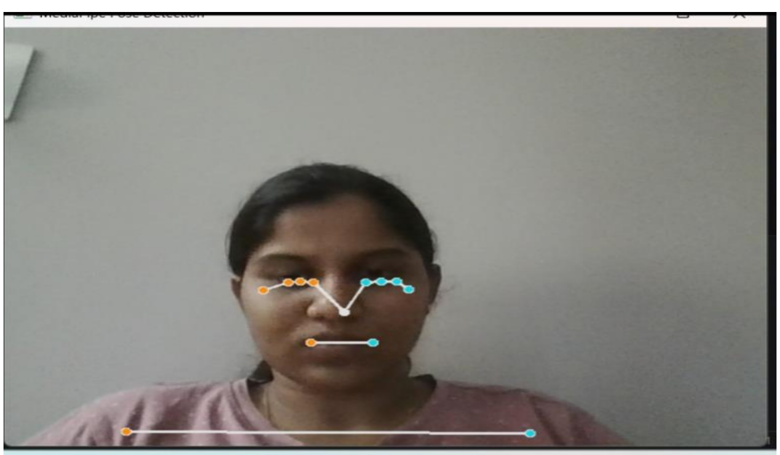
Task -1 Pose Detection Demo

Code:



```python
import cv2
import mediapipe as mp

# Initialize MediaPipe Pose
mp_pose = mp.solutions.pose
mp_drawing = mp.solutions.drawing_utils
mp_drawing_styles = mp.solutions.drawing_styles

pose = mp_pose.Pose(
    static_image_mode=False,
    model_complexity=1,
    smooth_landmarks=True,
    enable_segmentation=False,
    min_detection_confidence=0.5,
    min_tracking_confidence=0.5
)

# Open webcam
cap = cv2.VideoCapture(0)

if not cap.isOpened():
    print("Error: Could not open webcam.")
    exit()

print("Pose Detection Running... Press 'Q' to quit.")
```

Output:



Task 2- Shoulder Width Calculation

The horizontal distance across the body from the outer edge of one shoulder to the other. In clothing, this is the most critical measurement — if shoulder width is wrong, nothing fits properly.

A tape measure is placed across the back, from the bony point of one shoulder (acromion) to the other. Average: 40–44cm for men, 36–40cm for women.

MediaPipe gives landmark points 11 (left shoulder) and 12 (right shoulder). You calculate Euclidean distance between them in pixel space, then convert to cm using a calibration factor. The problem is these landmarks sit on the *joint*, not the outer edge of the shoulder, so results typically read 2–3cm narrower than a tape measure would.

Key observations

- Works best when person faces the camera directly — any rotation causes the width to appear smaller than it is
- Accuracy drops if arms are raised or crossed
- Lighting and loose clothing don't affect this one much since shoulders are bony and visible
- Most reliable of the three measurements

Limitations

- Z-axis (depth) is ignored — a person standing at an angle will give wrong results
- MediaPipe landmark sits on joint, not true shoulder edge
- Single-frame measurements are noisy — need to average across 10–15 frames

Height Estimation

Total body height from floor to crown of head. Used in fashion to determine dress/trouser length, sleeve length, and overall size category.

Person stands straight against a wall, flat feet, looking forward. A mark is made at the crown of the head. Simple but requires cooperation and a controlled environment.

MediaPipe doesn't landmark the top of the skull, so direct head-to-toe measurement isn't possible. Two main approaches:

1. **Nose-to-ankle span** — use LM 0 (nose) to LM 27/28 (ankles), add a fixed offset (~10cm) for the forehead. Simple but breaks with any body tilt.
2. **Torso proportion method** — the torso (shoulder midpoint to hip midpoint) is ~30% of height in adults. Measure that, divide by 0.30. More stable across different poses.

Key observations

- Torso method is more reliable but still depends on the 30% constant, which varies slightly by age and ethnicity
- Feet must be visible in frame — if cropped, height can't be estimated
- Works poorly if person is sitting, leaning, or wearing thick-soled shoes
- Camera must be at roughly chest height for minimal perspective distortion

Limitations

- Most error-prone of the three measurements (±5–10cm realistic)
- Perspective from camera angle causes significant over/underestimation
- Cannot detect whether person is standing straight or slouching

Waist Estimation

Circumference of the body at the narrowest point of the torso, typically 2–3cm above the navel. Critical for trousers, skirts, and fitted tops.

Tape measure wrapped around the body at the natural waistline. Person exhales normally — measurements taken while holding breath are often 3–5cm smaller than reality.

No MediaPipe landmark exists for the waist. Three approaches:

1. **Ratio method** — waist ≈ shoulder width × 0.84. Rough estimate only. Average body proportions assumed.
2. **Geometric interpolation** — the waist sits ~62% of the way from shoulder to hip vertically. Estimate width at that point using the silhouette.
3. **Silhouette scanning** — use background subtraction or segmentation to isolate the body, then scan horizontal rows in the torso area to find the narrowest pixel width. Most accurate without a depth camera.

Key observations

- Ratio method fails completely on non-average body shapes

- Silhouette method needs clean background — busy backgrounds break segmentation
- Waist is the hardest measurement to get right from a 2D image because it's a circumference, not a width — width × π assumes a circular cross-section which is never true
- Clothing makes this nearly useless — a baggy shirt hides the waist entirely

Limitations

- 2D image gives width, not circumference — major source of error
- Body shape (apple vs hourglass vs rectangle) changes the width-to-circumference ratio significantly
- Loose clothing adds 5–15cm of error with no way to correct it from image alone
- Requires segmentation which adds processing overhead and failure points

Task 3 — AI Research (Short Notes)

Virtual Try-On AI Systems

Virtual try-on systems use AI and computer vision to let users digitally wear clothes before buying them online.

1. OpenCV captures image/video
2. MediaPipe detects body landmarks
3. AI estimates body measurements
4. Clothes are aligned on the body
5. User sees virtual fitting result

Benefits:

- Better shopping experience
- Correct size selection
- Reduces product returns

Fashion Recommendation Model

A fashion recommendation model suggests clothes based on:

- user preferences
- shopping history
- body type
- fashion interests